

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281969

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KATO; Yusuke et al.

SAND MOLD-FORMING ADDITIVE, SAND MOLD-FORMING COMPOSITION, METHOD FOR PRODUCING SAND MOLD, AND SAND MOLD

Abstract

The sand mold-forming additive may contain a surfactant (A) including at least one selected from compounds represented by a formula (1) and compounds represented by a formula (8).

Inventors: KATO; Yusuke (Nagoya-shi, Aichi-ken, JP), ZENPO; Toshihiko (Nagoya-shi, Aichi-ken, JP), AOKI; Tomohiro (Nagoya-shi, Aichi-ken, JP), SATO; Shumma (Gamagori-shi, Aichi-ken, JP)

Applicant: SINTOKOGIO, LTD. (Nagoya-shi, Aichi, JP); TAKEMOTO YUSHI KABUSHIKI KAISHA (Gamagori-shi, Aichi, JP)

Family ID: 1000008671902

Assignee: SINTOKOGIO, LTD. (Nagoya-shi, Aichi, JP); TAKEMOTO YUSHI KABUSHIKI KAISHA (Gamagori-shi, Aichi, JP)

Appl. No.: 18/687471

Filed (or PCT Filed): September 22, 2022

PCT No.: PCT/JP2022/035374

Foreign Application Priority Data

JP	2021-161096	Sep. 30, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: B22C1/02 (20060101); B22C9/02 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a 35 U.S.C. § 371 U.S. National Phase entry of, and claims priority to, PCT Application PCT/JP2022/035374 filed Sep. 22, 2022, which claims priority to Japanese Patent Application No. 2021-161096 filed Sep. 30, 2021, which is hereby incorporated herein by reference in its entirety for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT [0002] Not applicable.

BACKGROUND

[0003] The present disclosure generally relates to sand mold-forming additives, sand mold-forming compositions, methods for producing sand molds, and sand molds.

[0004] Collapsible sand molds are conventionally used to manufacture castings with hollow sections. When manufacturing castings with hollow sections that have complex internal structures, the composition, which is the material for sand molds, needs high filling ability, especially fluidity. An increase in the water content in the composition can improve the fluidity. It is, however, undesirable because as the water content increases, the time required for water evaporation during sand mold production may increase and the strength of the sand mold may decrease. Thus, International Publication No. 2007/058254 discloses that the fluidity of the composition is improved by adding a surfactant such as polyglycerol fatty acid ester to the composition, which is the material of the mold, and then foaming it.

[0005] However, when producing castings with hollow sections having extremely complex internal structures, since materials for sand molds are required to have very high filling properties, conventional surfactants sometimes cannot sufficiently increase the fluidity of the composition. Accordingly, there has been a need for improved sand mold-forming additives.

SUMMARY

[0006] A first aspect of this disclosure is a sand mold-forming additive, comprising at least one surfactant (A) selected from compounds represented by a formula (1) below and compounds represented by a formula (8) below.

##STR00001##

(In the formula (1), R^{sup.1} is a hydrocarbon group with 6 to 22 carbon atoms, R^{sup.2} is a hydrogen atom, a hydrocarbon group with 1 to 4 carbon atom(s) or a hydroxyalkyl group with 1 to 4 carbon atom(s), and R^{sup.3} is a group represented by one of formulas (2) to (7) below.)

##STR00002##

(In the formulas (2) and (3), M^{sup.1} is an alkali metal atom; in the formulas (4) to (7), M^{sup.1} is an alkali metal atom or an organic amine, and in the formula (3), each of R^{sup.4} and R^{sup.5} is a hydrogen atom or a hydrocarbon group with 1 to 4 carbon atom(s).)

##STR00003##

(In the formula (8), R^{sup.6} is a hydrocarbon group with 6 to 22 carbon atoms, R^{sup.7} is an alkylene group with 2 to 4 carbon atoms, n is an integer from 2 to 5, and M^{sup.2} is an alkali metal atom.)

[0007] A second aspect of this disclosure is the sand mold-forming additive of the first aspect, wherein each of R^{sup.1} in the formula (1) and R^{sup.6} in the formula (8) is a hydrocarbon group with 8 to 14 carbon atoms.

[0008] A third aspect of this disclosure is the sand mold-forming additive of the first or second aspect, further comprising a surfactant (B) described below.

Surfactant (B): one selected from an alkyl sulfonate metal salt (B1), an alkyl sulfate metal salt (B2), a polyoxyalkylene alkyl ether sulfate metal salt (B3), an alkyl glycoside (B4), a polyoxyalkylene alkyl ether phosphate metal salt (B5), an alkyl benzene sulfonate metal salt (B6), an alkylamine (B7), an alkyl sulfosuccinate metal salt (B8), and a betaine surfactant (B9)

[0009] A fourth aspect of this disclosure is the sand mold-forming additive of the third aspect, wherein the surfactant (B) is the alkyl sulfonate metal salt (B1), the alkyl glycoside (B4), or the polyoxyalkylene alkyl ether phosphate metal salt (B5).

[0010] A fifth aspect of this disclosure is the sand mold-forming additive of the first or second aspect, further comprising a surfactant (B) described below.

Surfactant (B): at least two selected from an alkyl sulfonate metal salt (B1), an alkyl sulfate metal salt (B2), a polyoxyalkylene alkyl ether sulfate metal salt (B3), an alkyl glycoside (B4), a polyoxyalkylene alkyl ether phosphate metal salt (B5), an alkyl benzene sulfonate metal salt (B6), an alkylamine (B7), an alkyl sulfosuccinate metal salt (B8), and a betaine surfactant (B9)

[0011] A sixth aspect of this disclosure is the sand mold-forming additive of the fifth aspect, wherein one of the surfactant (B) is the alkyl sulfonate metal salt (B1) or the alkyl glycoside (B4).

[0012] A seventh aspect of this disclosure is the sand mold-forming additive of any one of the third to sixth aspects, wherein the surfactant (A) is 1 to 40 parts by mass and the surfactant (B) is 60 to 99 parts by mass when a total content of the surfactant (A) and the surfactant (B) is 100 parts by mass.

[0013] An eighth aspect of this disclosure is a sand mold-forming composition, comprising the sand mold-forming additive of any one of the first to seventh aspects, sand, and a silicate compound (C) below.

Silicate compound (C): at least one selected from sodium silicate, potassium silicate, and lithium silicate

[0014] A ninth aspect of this disclosure is the sand mold-forming composition of the eighth aspect, wherein the sand mold-forming additive is 1 to 100 parts by mass with respect to 100 parts by mass of the silicate compound (C).

[0015] A tenth aspect of this disclosure is a method for producing a sand mold, comprising a process A for molding the sand mold-forming composition of the eighth or ninth aspect.

[0016] An eleventh aspect of this disclosure is the method for producing the sand mold of the tenth aspect, wherein the process A comprises a step A1 for stirring and forming the sand mold-forming composition of the eighth or ninth aspect, a step A2 for filling a molding space with the sand mold-forming composition in a foamed state, and a step A3 for solidifying the filled sand mold-forming composition.

[0017] A twelfth aspect of this disclosure is a method for producing a sand mold, comprising a step B1 for preparing the sand mold-forming composition of the eighth or ninth aspect by mixing and foaming the sand mold-forming additive of any one of the first to seventh aspects and the silicate compound (C) and then adding the sand thereto, a step B2 for filling a molding space with the prepared sand mold-forming composition, and a step B3 for solidifying the filled sand mold-forming composition.

[0018] A thirteenth aspect of this disclosure is a sand mold, comprising the sand mold-forming additive of any one of the first to seventh aspects, sand, and a silicate compound (C) below.

Silicate compound (C): at least one selected from sodium silicate, potassium silicate, and lithium silicate

[0019] The numerical range represented by "XX-YY" herein means the range including its upper and lower limits. In other words, the term "XX-YY" means "XX or more and YY or less."

[0020] According to the present disclosure, it is possible to provide a sand mold-forming additive that produces a sand mold-forming composition with excellent flowability. In addition, the sand

mold-forming composition, a sand mold, and a method for producing sand molds using the sand mold-forming additive can be provided.

Description

DETAILED DESCRIPTION

<Sand Mold-Forming Additive>

[0021] The sand mold-forming additive contains a surfactant (A). The sand mold-forming additive also optionally contains a surfactant (B) as an auxiliary agent for the surfactant (A) that is a foaming agent.

[0022] The sand mold-forming additive is mixed with sand together with a silicate compound (C), which is a binder, to prepare a sand mold-forming composition, which is the material for sand molds.

<Surfactant (A)>

[0023] The surfactant (A) corresponds to compounds represented by the formula (1) below and compounds represented by the formula (8) below. As the surfactant (A), a single compound may be used alone, and two or more may be used in combination.

##STR00004##

(In the formula (1), R.sup.1 is a hydrocarbon group with 6 to 22 carbon atoms, R.sup.2 is a hydrogen atom, a hydrocarbon group with 1 to 4 carbon atom(s) or a hydroxyalkyl group with 1 to 4 carbon atom(s), and R.sup.3 is a group represented by one of the formulas (2) to (7) below.)

##STR00005##

(In the formulas (2) and (3), M.sup.1 is an alkali metal atom; in the formulas (4) through (7), M.sup.1 an alkali metal atom or an organic amine; and in the formula (3), each of R.sup.4 and R.sup.5 is a hydrogen atom or a hydrocarbon group with 1 to 4 carbon atom(s).)

##STR00006##

(In the formula (8), R.sup.6 is a hydrocarbon group with 6 to 22 carbon atoms, R.sup.7 is an alkylene group with 2 to 4 carbon atoms, n is an integer from 2 to 5, and M.sup.2 is an alkali metal atom.)

[0024] The hydrocarbon group with 6 to 22 carbon atoms, corresponding to each of R.sup.1 in the formula (1) and R.sup.6 in the formula (8), may include a linear hydrocarbon group, a hydrocarbon group with side chain(s), and a cyclic hydrocarbon group. Each of R.sup.1 in the formula (1) and R.sup.6 in the formula (8) is preferably a hydrocarbon group with 8 to 14 carbon atoms and more preferably an alkyl group with 8 to 14 carbon atoms from the viewpoint of the strength of the resulting sand molds, etc.

[0025] Examples of the alkali metal atom, corresponding to M.sup.1 in the formulas (2) through (7) and M.sup.2 in the formula (8), include sodium, potassium, and lithium, etc. The organic amine, corresponding to M.sup.1 in the formulas (4) through (7), includes alkanolamines each having a positive charge in the molecule as represented by a formula (11). Examples of such alkanolamines include monoethanolamine, diethanolamine, and triethanolamine.

##STR00007##

(In the formula (11), each of R.sup.10, R.sup.11, and R.sup.12 is a hydroxyalkyl group with 1 to 4 carbon atom(s) or hydrogen atom such that at least one of R.sup.10, R.sup.11, and R.sup.12 is a hydroxyalkyl group with 1 to 4 carbon atom(s).)

<Surfactant (A)>

[0026] The surfactant (A) includes amino acid-based surfactants represented by the formula (1) and/or alkyl ether carboxylates. The amino acid-based surfactants may include N-acyl taurine salt type anionic surfactants such as sodium cocoyl methyl taurate, sodium lauroyl methyl taurate, sodium myristoyl methyl taurate, sodium palmitoyl methyl taurate, sodium stearoyl methyl taurate,

and sodium taurine cocoyl methyl taurate; N-acylamino acid salt type anionic surfactants (alanine-based surfactants) such as sodium cocoyl alaninate, sodium cocoyl methyl alaninate, sodium lauroyl methyl alaninate, sodium myristoyl methyl alaninate, sodium lauroyl hydroxyethyl- β -alaninate; carboxylate-type anionic surfactants such as alkyl ether carboxylates derived from ethylene oxide adducts of higher alcohols (more specifically, sodium laureth-4 carboxylate, sodium laureth-6 carboxylate, etc.), and alkyl glucose carboxylates (more specifically, sodium cocoyl glucose carboxylate, sodium lauryl glucose carboxylate, etc.); glutamic acid-based surfactants such as sodium cocoyl glutamate, sodium lauroyl glutamate, and sodium myristoyl glutamate, sarcosine-based surfactants such as sodium cocoyl sarcosinate, potassium cocoyl sarcosinate, cocoyl sarcosine triethanolamine salt, sodium lauroyl sarcosinate, potassium lauroyl sarcosinate, lauroyl sarcosine triethanolamine salt, and sodium myristoyl sarcosinate; glycine-based surfactants such as sodium cocoyl glycinate, potassium cocoyl glycinate, and cocoyl glycine triethanolamine salt; and aspartic acid-based surfactants such as sodium cocoyl aspartate and sodium lauroyl aspartate.

<Surfactant (B)>

[0027] The surfactant (B) includes at least one selected from an alkyl sulfonate metal salt (B1), an alkyl sulfate metal salt (B2), a polyoxyalkylene alkyl ether sulfate metal salt (B3), an alkyl glycoside (B4), a polyoxyalkylene alkyl ether phosphate metal salt (B5), an alkylbenzenesulfonate metal salt (B6), an alkylamine (B7), an alkyl sulfosuccinate metal salt (B8), and a betaine surfactant (B9). The surfactant (B) may improve the flowability of the sand mold-forming composition. As the surfactant (B), a single compound can be used alone, however two or more are preferably used together to increase the strength of the sand mold.

[0028] When the sand mold-forming additive contains a single compound as the surfactant (B), the surfactant (B) is preferably the alkyl sulfonate metal salt (B1), the alkyl glycoside (B4), or the polyoxyalkylene alkyl ether phosphate metal salt (B5) from the viewpoint of the fluidity of the sand mold-forming composition.

[0029] When the sand mold-forming additive contains two or more compounds as the surfactant (B), one of the compounds as the surfactant (B) is preferably the alkyl sulfonate metal salt (B1) or the alkyl glycoside (B4) from the viewpoint of the flowability of the sand mold-forming composition. The alkyl sulfonate metal salt (B1) and the alkyl glycoside (B4) may be used together as the surfactant (B).

<Alkyl sulfonate metal salt (B1)>

[0030] Known alkyl sulfonate metal salts can be used as the alkyl sulfonate metal salt (B1). As the alkyl sulfonate metal salt (B1), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkyl sulfonate metal salt (B1) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkyl sulfonate metal salt (B1) may have a linear or branched structure. The metal of the alkyl sulfonate metal salt (B1) may be alkali metal such as lithium, sodium, and potassium, or alkaline earth metal such as calcium and magnesium, and is preferably sodium.

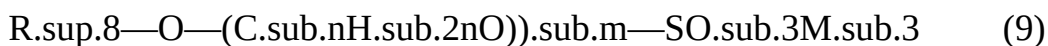
<Alkyl sulfate metal salt (B2)>

[0031] Known alkyl sulfate metal salts can be used as the alkyl sulfate metal salt (B2). As the alkyl metal sulfate (B2), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkyl metal sulfate (B2) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkyl metal sulfate (B2) may have a linear or branched structure. The metal of the alkyl sulfate metal salt (B2) may be alkali metal such as lithium, sodium, and potassium, or alkaline earth metal such as calcium and magnesium, and is preferably sodium.

<Polyoxyalkylene alkyl ether sulfate metal salt (B3)>

[0032] Known polyoxyalkylene alkyl ether sulfate metal salts can be used as the polyoxyalkylene alkyl ether sulfate metal salt (B3). As the polyoxyalkylene alkyl ether sulfate metal salt (B3), a single compound may be used alone, and two or more may be used in combination. The carbon

number of the alkyl group of the polyoxyalkylene alkyl ether sulfate metal salt (B3) is preferably 8 to 16. The structure and the molar number of the alkylene oxide adduct of the polyoxyalkylene alkyl ether sulfate metal salt (B3) are not particularly limited. The metal of the polyoxyalkylene alkyl ether sulfate metal salt (B3) may be alkali metal such as lithium, sodium, and potassium, or alkaline earth metal such as calcium and magnesium, and is preferably sodium.
[0033] For example, compounds represented by a formula (9) below can be used as the polyoxyalkylene alkyl ether sulfate metal salt (B3).



(In the formula (9), R.sup.8 is an alkyl group with 8 to 16 carbon atoms, n is an integer from 2 to 4, m is 1 to 10, and M.sub.3 is an alkali metal.)

In the formula (9), the carbon number of R.sup.8 is preferably 10 to 14, and R8 may have a linear or branched structure. In the formula (9), n is preferably 2 or 3, and 2 is more preferred. In the formula (9), m is the average number of added moles in the polyoxyalkylene group, and is preferably 1 to 6, more preferably 1 to 4. Examples of M.sub.3 in the formula (9) include lithium, sodium, potassium, etc., with sodium being preferred.

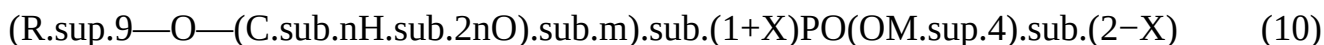
<Alkyl glycoside (B4)>

[0034] Known alkyl glycosides can be used as the alkyl glycoside (B4). As the alkyl glycoside (B4), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkyl glycoside (B4) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkyl glycoside (B4) may have a linear or branched structure. Examples of the alkyl glycoside (B4) include alkyl glucosides and alkyl maltosides, in which a sugar and a higher alcohol are glycosidically bonded to each other, as well as alkyl thioglucosides and alkyl thiomaltosides, in which a sugar and a higher alcohol are thioglucosidically bonded to each other.

<Polyoxyalkylene alkyl ether phosphate metal salt (B5)>

[0035] Known polyoxyalkylene alkyl ether phosphate metal salts can be used as the polyoxyalkylene alkyl ether phosphate metal salt (B5). As the polyoxyalkylene alkyl ether phosphate metal salt (B5), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the polyoxyalkylene alkyl ether phosphate metal salt (B5) is preferably 8 to 22. The structure and the molar number of the alkylene oxide adduct of the polyoxyalkylene alkyl ether phosphate metal salt (B5) are not particularly limited. The polyoxyalkylene alkyl ether phosphate metal salt (B5) may be a phosphate monoester or a phosphate diester. The metal of the polyoxyalkylene alkyl ether phosphate metal salt (B5) may be an alkali metal such as lithium, sodium, and potassium, or an alkaline earth metal such as calcium and magnesium and is preferably sodium.

[0036] For example, compounds represented by a formula (10) below can be used as the polyoxyalkylene alkyl ether phosphate metal salt (B5).



(In the formula (10), R.sup.9 is an alkyl group with 8 to 22 carbon atoms, n is an integer from 2 to 4, m is 1 to 10, X is 0 or 1, and M.sub.4 is an alkali metal.)

In the formula (10), the carbon number of R.sup.9 is preferably 8 to 18. R.sup.9 may have a linear or branched structure. In the formula (10), n is preferably 2 or 3, and more preferably 2. In the formula (10), m is the average number of added moles in the polyoxyalkylene group and is preferably 1 to 6, more preferably 1 to 4. Examples of M.sub.4 in the formula (10) include lithium, sodium, potassium, etc., and sodium is preferred. When X is 1, a pair of (R.sup.9—O—(C.sub.nH.sub.2O).sub.m) groups may be mutually identical or different from each other. When X is 0, one of two (OM.sub.4) groups may be replaced with a hydroxyl group.

<Alkylbenzene sulfonate metal salt (B6)>

[0037] Known alkylbenzene sulfonate metal salts can be used as the alkylbenzene sulfonate metal salt (B6). As the alkylbenzene sulfonate metal salt (B6), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkylbenzene sulfonate metal salt (B6) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkylbenzene sulfonate metal salt (B6) may have a linear or branched structure. The metal of the alkylbenzene sulfonate metal salt (B6) may be alkali metal such as lithium, sodium, and potassium, or alkaline earth metal such as calcium and magnesium, and is preferably sodium.

<Alkylamine (B7)>

[0038] Known alkylamine oxides and polyoxyalkylene alkylamines can be used as the alkylamine (B7). As the alkylamine (B7), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkylamine (B7) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkylamine (B7) may have a linear or branched structure. When the alkylamine (B7) is a polyoxyalkylene alkylamine, the structure and the molar number of the alkylene oxide adduct are not particularly limited.

<Alkyl sulfosuccinate metal salt (B8)>

[0039] Known alkyl sulfosuccinate metal salts can be used as the alkyl sulfosuccinate metal salt (B8). As the alkyl sulfosuccinate metal salt (B8), a single compound may be used alone, and two or more may be used in combination. The carbon number of the alkyl group of the alkyl sulfosuccinate metal salt (B8) is preferably 8 to 22, and more preferably 8 to 18. The alkyl group of the alkyl sulfosuccinate metal salt (B8) may have a linear or branched structure. The metal of the alkyl sulfosuccinate metal salt (B8) may be alkali metal such as lithium, sodium, and potassium, or alkaline earth metal such as calcium and magnesium, and is preferably sodium.

<Betaine-based Surfactant (B9)>

[0040] Known surfactants each having a betaine structure can be used as the betaine-based surfactant (B9). As the betaine-based surfactant (B9), a single compound may be used alone, and two or more may be used in combination. Examples of the betaine-based surfactant (B9) may include alkyl betaine surfactants, alkyl amidopropyl betaine surfactants, alkyl carboxymethyl hydroxyethyl imidazolinium betaine surfactants, alkyl hydroxysulfobetaine surfactants, and alkyl dimethylamino acetate betaine surfactants, etc. The carbon number of the alkyl group of the betaine-based surfactant (B9) is preferably 8 to 22, and more preferably 10 to 18.

<Mixing Ratio of Sand Mold-Forming Additive>

[0041] The sand mold-forming additive preferably contains 1 to 40 parts by mass of the surfactant (A) and 60 to 99 parts by mass of the surfactant (B) based on 100 parts by mass of the total content of the surfactant (A) and the surfactant (B). When the content ratio of the surfactants (A) and (B) is within the above range, it is possible to provide sand mold-forming compositions with excellent flowability and sand molds with sufficient strength.

<Sand Mold-Forming Composition>

[0042] The sand mold-forming composition contains the sand mold-forming additive, sand, and a silicate compound (C) as a binder. The silicate compound (C) is selected from sodium silicate, potassium silicate, and lithium silicate. As the silicate compound (C), a single compound may be used alone, or two or more may be used in combination. The silicate compound (C) may be used either in solid form or in liquid form such as water glass.

[0043] The sand for the sand mold-forming composition is not particularly limited, and sand grains that are conventionally known aggregates can be used therefor. For example, silica sand, alumina sand, olivine sand, chromite sand, zircon sand, mullite sand, various types of artificial sand (so-called artificial aggregate), and recycled sand may be used.

[0044] The grain size of the sand is not limited. A grain size index (AFS) is preferably 3 to 300, and is more preferably 20 to 200. When the grain size index is 3 or higher, the sand mold-forming composition may have excellent flowability to improve filling property during production of sand molds. When the grain size index is 300 or less, the sand molds may maintain good air

permeability.

[0045] The sand can have any shape without being particularly limited. Particularly, spherical shape is preferred. When the shape is spherical, the sand mold-forming composition may have excellent fluidity to improve the filling property during production of sand molds.

<Mixing Ratio of Sand Mold-Forming Composition>

[0046] The sand mold-forming composition can contain sand, the silicate compound (C) and the sand mold-forming additive in any ratio. The sand mold-forming composition contains preferably 1 to 100 parts by mass, more preferably 1 to 10 parts by mass of the sand mold-forming additive, based on 100 parts by mass of the silicate compound (C). The sand mold-forming composition contains preferably 1,000 to 100,000 parts by mass, more preferably 4,000 to 70,000 parts by mass, even more preferably 7,000 to 30,000 parts by mass of the sand, with respect to 100 parts by mass of the silicate compound (C).

<Other Additives>

[0047] The sand mold-forming composition may preferably contain a liquid medium to improve fluidity and foamability. The liquid medium is only required to be in liquid form. Conventionally known liquid mediums can be used without being particularly limited. Examples of the liquid medium include water, ethylene glycol, propylene glycol, etc. Water is preferred from the perspectives of cost, ease of liquid medium removal, and safety. The content ratio of the liquid medium may be set appropriately according to the type of the sand and the content ratio of each component. Even when the sand mold-forming composition contains an excessive amount of the liquid medium, the fluidity of the sand mold-forming composition increases, but it is undesirable because the time for solidification of the sand mold-forming composition during the production of sand molds increases and the strength of the sand mold may decrease.

[0048] Other conventionally known additives can be added to the sand mold-forming composition according to the purpose. For example, such additives include hardeners that accelerate the solidification of the sand mold-forming composition during the production of sand molds.

<Sand Mold>

[0049] The sand mold is made by molding the sand mold-forming composition and contains the sand mold-forming additive, the silicate compound (C), and the sand.

<Method for Producing Sand Mold>

[0050] The sand mold-forming composition is produced by mixing the above-described ingredients with each other. The order and method of mixing are not particularly limited. For example, the sand mold-forming additive and the silicate compound (C) may be mixed and then added to the sand. Alternatively, all the materials may be mixed simultaneously. Conventionally known mixing devices can be used as a mixing device for blending and stirring the sand with the sand mold-forming additive and the silicate compound (C), without being particularly limited. Either a batch-type mixing device or a continuous-type mixing device can be used.

[0051] The molding of the sand mold-forming composition may be done by hand or by a molding machine. Conventional molding machines can be used as the molding machine, without being particularly limited. Examples of the molding machine include jolt molding machines, jolt squeeze molding machines, blow squeeze molding machines, static pressure molding machines, core molding machines, etc.

[0052] For filling the metal mold (permanent mold) with the sand mold-forming composition, it is preferable to press the sand mold-forming composition in a foamed state (hereinafter referred to as foamed composition) into the molding space in the heated metal mold, and it is more preferable to inject the composition into the metal mold.

[0053] The methods of filling the molding space with the foamed composition include direct pressurization with a piston inside the cylinder, filling by supplying compressed air into the cylinder, pressure-feeding with a screw, and pouring. The direct pressurization with a piston and filling with compressed air are preferred from the viewpoint of filling speed and filling stability due

to uniform pressurization of the foamed composition.

[0054] The method of solidifying the foamed composition filled in the molding space of the metal mold is not particularly limited. It is preferable to evaporate liquid, including the liquid medium, of the foamed composition. Evaporation of the liquid, including the liquid medium, of the foamed composition may be performed by heat transfer from the metal mold to the foamed composition, by flow of hot air into the molding space, or by a combination thereof.

[0055] Specifically, a sand mold can be produced by a method including a process A for molding the sand mold-forming composition. The process A preferably includes the steps A1 to A4 below.

[0056] Step A1: Mixing a sand mold-forming additive, a silicate compound (C), a liquid medium, and sand with each other to prepare a sand mold-forming composition, and then stirring the sand

mold-forming composition to foam the same [0057] Step A2: Filling a molding space of a metal mold with the sand mold-forming composition in a foamed state (foamed composition) [0058] Step

A3: Evaporating liquid, including the liquid medium, from the sand mold-forming composition in the molding space to solidify the sand mold-forming composition [0059] Step A4: Removing the solidified sand mold-forming composition (sand mold) from the molding space

[0060] The sand mold can also be produced by a method that includes the steps B1 through B4

below. [0061] Step B1: Foaming a mixture of a sand mold-forming additive, a silicate compound (C) and a liquid medium, and then mixing it with sand to prepare a foamed sand mold-forming composition (foamed composition) [0062] Step B2: Filling a molding space of a metal mold with the sand mold-forming composition (foamed composition) [0063] Step B3: Evaporating liquid,

including the liquid medium, from the sand mold-forming composition in the molding space to solidify the sand mold-forming composition [0064] Step B4: Removing the solidified sand mold-forming composition (sand mold) from the molding space

[0065] In the step B1, the mixture of the sand mold-forming additive and the liquid medium may be foamed and then mixed with the silicate compound (C) and the sand.

[0066] In the step B1, the method of foaming the mixture of the sand mold-forming additive, the silicate compound (C) and the liquid medium is not particularly limited. For example, it is possible to utilize a method for supplying gas into the liquid to uniformly mix and disperse the liquid by fluid shear, etc., or a method for dissolving gas in the liquid to saturation under pressure and then rapidly decompressing to generate bubbles.

[0067] In order to improve the filling property of the foamed composition into the molding space, the viscosity of the foamed composition when filling the molding space is preferably 0.50 to 3.58 Pa·s, and is more preferably 0.50 to 1.12 Pa·s.

[0068] During the specific step described above or an interval between such steps, it is also possible to temporarily (e.g., for a week) store an intermediate product in a sealed container or the like. For example, in the step A1, the mixture of the sand mold-forming additive, the silicate compound (C) and the liquid medium may be temporarily stored and then mixed with sand to prepare the sand mold-forming composition. The foamed composition obtained in the step A1 may be temporarily stored, and the sand mold-forming composition may be stirred until it becomes in a foamed state again before the step A2. In the step B1, the mixture of the sand mold-forming additive, the silicate compound (C) and the liquid medium may be temporarily stored without being foamed, and then may be foamed.

EXAMPLE

[0069] Next, the present disclosure will be described more specifically with examples and comparative examples. The present disclosure is not limited to these.

<Preparation of Sand Mold-Forming Additive>

[0070] 5.00 parts by mass of sodium taurine octyl methyl taurate (A-1), 5.00 parts by mass of sodium decyl methyl taurate (A-2), 80.00 parts by mass of sodium alkyl sulfonate having an alkyl group with 8 to 14 carbon atoms (B-1), and 10.00 parts by mass of sodium alkyl sulfate having an alkyl group with 8 to 14 carbon atoms (B-2) were uniformly mixed to prepare a sand mold-forming

additive (H-1).

[0071] Sand mold-forming additives (H-2 to H-39, h-1 to h-2) having component compositions shown in Tables 1 and 2 below were prepared in the same manner as the sand mold-forming additive (H-1).

TABLE-US-00001 TABLE 1 Sand Mold-Forming Additive Components																			
Surfactant (B) Other					Surfactant (A)														
B-1	B-2	B-3	B-4	B-5	Surfactants Name	Kind	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Kind
H-1	A-1	5.00	80.00	10.00	—	—	—	—	—	A-2	5.00	H-2	A-7	5.00	80.00	10.00	—	—	—
A-2	5.00	H-3	A-1	25.00	60.00	—	10.00	—	—	—	A-2	5.00	H-4	A-7	25.00	60.00	—	10.00	—
—	—	A-8	5.00	H-5	A-1	10.00	40.00	—	—	40.00	—	—	A-2	10.00	H-6	A-8	10.00	40.00	—
40.00	—	—	A-2	10.00	H-7	A-1	8.00	30.00	30.00	—	—	30.00	—	—	A-2	2.00	H-8	A-1	8.00
30.00	30.00	—	—	30.00	—	—	A-9	2.00	H-9	A-9	8.00	30.00	30.00	—	—	30.00	—	A-2	2.00
H-10	A-3	5.00	—	30.00	—	40.00	—	—	—	A-2	25.00	H-11	A-9	5.00	—	30.00	—	40.00	—
25.00	H-12	A-3	5.00	—	—	40.00	40.00	10.00	—	—	A-2	5.00	H-13	A-1	2.00	—	—	80.00	10.00
—	—	A-2	8.00	H-14	A-1	2.00	—	—	—	80.00	10.00	—	—	A-8	8.00	H-15	A-1	10.00	80.00
—	—	—	—	H-16	A-7	10.00	80.00	10.00	—	—	—	—	—	H-17	A-8	10.00	80.00	10.00	—
—	H-18	A-2	20.00	40.00	—	—	40.00	—	—	—	H-19	A-9	20.00	40.00	—	—	40.00	—	—

TABLE-US-00002 TABLE 2 Sand Mold-Forming Additive Components																						
Surfactant (A)		B-1	B-2	B-3	B-4	B-5	Surfactants Name		Kind	Parts	Parts	Parts	Parts	Parts	Parts	Parts	Kind					
Parts	H-20	A-3	10.00	30.00	10.00	10.00	—	40.00	—	—	H-21	A-1	30.00	—	40.00	—	30.00	—				
—	H-22	A-1	10.00	—	30.00	60.00	—	—	—	H-23	A-1	10.00	—	60.00	—	20.00	—	A-4				
10.00	H-24	A-1	10.00	—	60.00	—	—	20.00	—	—	A-7	10.00	H-25	A-1	10.00	—	60.00	—	20.00			
—	—	A-9	10.00	H-26	A-1	20.00	—	60.00	—	—	20.00	—	H-27	A-1	25.00	—	20.00	—	50.00			
—	—	A-2	5.00	H-28	A-1	5.00	90.00	—	—	—	—	—	A-2	5.00	H-29	A-1	10.00	—	—	70.00	—	
—	—	A-2	20.00	H-30	A-2	30.00	—	—	—	70.00	—	—	H-31	A-1	10.00	—	—	—	80.00	—	—	
A-2	10.00	H-32	A-1	25.00	—	70.00	—	—	—	—	A-2	5.00	H-33	A-1	5.00	—	—	70.00	—	—	—	
—	A-2	25.00	H-34	A-5	5.00	90.00	—	—	—	—	—	A-6	5.00	H-35	A-5	10.00	—	—	70.00	—	—	
—	—	A-6	20.00	H-36	A-1	100.00	—	—	—	—	—	—	H-37	A-2	100.00	—	—	—	—	—	H-38	
A-5	100.00	—	—	—	—	—	—	—	—	H-39	A-6	100.00	—	—	—	—	—	—	h-1	—	—	—
—	b1	100.00	h-2	—	—	—	—	—	—	—	b2	100.00	—	—	—	—	—	—	—	—	—	

[0072] The components shown in Tables 1 and 2 are as follows.

[Surfactant (A)]

[0073] A-1: Sodium taurine octyl methyl taurate [0074] A-2: Sodium decyl methyl taurate [0075] A-3: Sodium taurine alkyl methyl taurate having an alkyl group with 8 to 14 carbon atoms [0076] A-4: Potassium alkyl methyl taurate having an alkyl group with 8 to 14 carbon atoms [0077] A-5: Sodium taurine alkyl methyl taurate having an alkyl group with 16 to 18 carbon atoms [0078] A-6: Sodium alkyl methyl taurate having an alkyl group with 16 to 18 carbon atoms [0079] A-7: Triethanolamine cocoyl sarcosinate having an alkyl group with 8 to 14 carbon atoms [0080] A-8: Sodium cocoyl sarcosinate having an alkyl group with 8 to 14 carbon atoms [0081] A-9: Sodium lauroyl hydroxyethyl- β -alaninate

[Surfactant (B)]

[0082] B-1: Sodium alkyl sulfonate having an alkyl group with 8 to 14 carbon atoms [0083] B-2: Sodium alkyl sulfate having an alkyl group with 8 to 14 carbon atoms [0084] B-3: Sodium polyoxyethylene alkyl ether sulfate, which has an alkyl group with 10 to 14 carbon atoms and in which the number of moles of added ethylene oxides is 1 to 4 moles [0085] B-4: n-Alkyl-B-D-glucoside having an alkyl group with 8 to 14 carbon atoms [0086] B-5: Sodium polyoxyethylene alkyl ether phosphate, which has an alkyl group with 8 to 18 carbon atoms and in which the number of moles of added ethylene oxides is 1 to 4 moles

[Other Surfactants]

[0087] b1: Diglyceryl monolaurate [0088] b2: Sorbitan monolaurate

Example 1

(Evaluation Criteria for Fluidity)

[0100] 5:0.50 to 1.12 [Pa.Math.s] [0101] 4:1.13 to 1.34 [Pa.Math.s] [0102] 3:1.35 to 1.50

[Pa.Math.s] [0103] 2:1.51 to 3.58 [Pa.Math.s] [0104] 1:3.59 or more [Pa.Math.s]

<Sand Mold Strength>

[0105] The sand mold-forming composition in a foamed state was injected into a metal mold heated to 250° C. by a molding machine (manufactured by SINTOKOGIO, LTD.). The metal mold is a mold for making a flexural strength test piece and has a cavity with a capacity of about 80,000 mm³. Injection was performed at a gate speed of 1 m/second and a cylinder surface pressure of 0.4 MPa. The sand mold-forming composition filled in the heated metal mold was left for 2 minutes to allow the heat of the metal mold to evaporate liquid, thereby solidifying the sand mold-forming composition. After solidification was completed, the sand mold was removed from the metal mold. A flexural strength test piece being 10 mm by 10 mm by 70 mm was made from the obtained sand mold. The flexural strength test piece was humidified at 20° C. and 60% humidity for 1 hour, and then the flexural strength (MPa) was measured. The flexural strength was measured in accordance with JACT Test Method SM-1, Flexural Strength Test Method. Sand mold strength was evaluated based on the measured flexural strength according to the following criteria. (Evaluation Criteria for Sand Mold Strength)

[0106] 5:3.51 to 4.00 MPa [0107] 4:3.26 to 3.50 MPa [0108] 3:3.01 to 3.25 MPa [0109] 2:2.50 to 3.00 MPa [0110] 1:2.49 MPa or less

TABLE-US-00004 TABLE 4 Sand Mold-Forming Results Composition (Name) Fluidity Sand Mold Strength Example 1 N-1 5 5 Example 2 N-2 5 5 Example 3 N-3 5 5 Example 4 N-4 5 5 Example 5 N-5 5 5 Example 6 N-6 5 5 Example 7 N-7 5 5 Example 8 N-8 5 5 Example 9 N-9 5 5 Example 10 N-10 5 5 Example 11 N-11 5 5 Example 12 N-12 5 5 Example 13 N-13 5 5 Example 14 N-14 5 5 Example 15 N-15 5 5 Example 16 N-16 5 5 Example 17 N-17 5 5 Example 18 N-18 5 5 Example 19 N-19 5 5 Example 20 N-20 5 5 Example 21 N-21 5 5 Example 22 N-22 5 5 Example 23 N-23 4 5 Example 24 N-24 4 5 Example 25 N-25 4 5 Example 26 N-26 4 5 Example 27 N-27 4 5 Example 28 N-28 4 4 Example 29 N-29 4 4 Example 30 N-30 4 4 Example 31 N-31 4 4 Example 32 N-32 3 4 Example 33 N-33 3 4 Example 34 N-34 3 3 Example 35 N-35 3 3 Example 36 N-36 2 3 Example 37 N-37 2 3 Example 38 N-38 2 2 Example 39 N-39 2 2 Comparative n-1 1 2 Example 1 Comparative n-2 1 2 Example 2

[0111] In Examples 1 to 39, the sand mold-forming compositions had excellent fluidity, and sand molds with sufficient strength could be obtained. In contrast, Comparative Examples 1 and 2 had low flowability because they did not contain surfactant (A).

Claims

1. A sand mold-forming additive, comprising: a surfactant (A) including at least one selected from compounds represented by a formula (1) below and compounds represented by a formula (8) below: ##STR00008## (In the formula (1), R^{sup.1} is a hydrocarbon group with 6 to 22 carbon atoms, R^{sup.2} is a hydrogen atom, a hydrocarbon group with 1 to 4 carbon atom(s) or a hydroxyalkyl group with 1 to 4 carbon atom(s), and R^{sup.3} is a group represented by one of formulas (2) to (7) below.) ##STR00009## (In the formulas (2) and (3), M^{sup.1} is an alkali metal atom; in the formulas (4) to (7), M^{sup.1} is an alkali metal atom or an organic amine; and in the formula (3), each of R^{sup.4} and R^{sup.5} is a hydrogen atom or a hydrocarbon group with 1 to 4 carbon atom(s).) ##STR00010## (In the formula (8), R^{sup.6} is a hydrocarbon group with 6 to 22 carbon atoms, R^{sup.7} is an alkylene group with 2 to 4 carbon atoms, n is an integer from 2 to 5, and M^{sup.2} is an alkali metal atom.)

2. The sand mold-forming additive of claim 1, wherein each of R^{sup.1} in the formula (1) and R^{sup.6} in the formula (8) is a hydrocarbon group with 8 to 14 carbon atoms.

3. The sand mold-forming additive of claim 1, further comprising a surfactant (B), wherein the

- surfactant (B) comprise one selected from an alkyl sulfonate metal salt (B1), an alkyl sulfate metal salt (B2), a polyoxyalkylene alkyl ether sulfate metal salt (B3), an alkyl glycoside (B4), a polyoxyalkylene alkyl ether phosphate metal salt (B5), an alkyl benzene sulfonate metal salt (B6), an alkylamine (B7), an alkyl sulfosuccinate metal salt (B8), and a betaine surfactant (B9).
- 4.** The sand mold-forming additive of claim 3, wherein the surfactant (B) is the alkyl sulfonate metal salt (B1), the alkyl glycoside (B4), or the polyoxyalkylene alkyl ether phosphate metal salt (B5).
- 5.** The sand mold-forming additive of claim 1, further comprising a surfactant (B) wherein the surfactant (B) comprise at least two selected from an alkyl sulfonate metal salt (B1), an alkyl sulfate metal salt (B2), a polyoxyalkylene alkyl ether sulfate metal salt (B3), an alkyl glycoside (B4), a polyoxyalkylene alkyl ether phosphate metal salt (B5), an alkyl benzene sulfonate metal salt (B6), an alkylamine (B7), an alkyl sulfosuccinate metal salt (B8), and a betaine surfactant (B9).
- 6.** The sand mold-forming additive of claim 5, wherein one of the at least two is the alkyl sulfonate metal salt (B1) or the alkyl glycoside (B4).
- 7.** The sand mold-forming additive of claim 3, wherein the surfactant (A) is 1 to 40 parts by mass and the surfactant (B) is 60 to 99 parts by mass when a total content of the surfactant (A) and the surfactant (B) is 100 parts by mass.
- 8.** A sand mold-forming composition, comprising the sand mold-forming additive of claim 1, sand, and a silicate compound (C), wherein the silicate compound (C) comprise at least one selected from sodium silicate, potassium silicate, and lithium silicate
- 9.** The sand mold-forming composition of claim 8, wherein the sand mold-forming additive is 1 to 100 parts by mass with respect to 100 parts by mass of the silicate compound (C).
- 10.** A method for producing a sand mold, comprising a process A for molding the sand mold-forming composition of claim 8.
- 11.** The method for producing the sand mold of claim 10, wherein the process A comprises: a step A1 for stirring and foaming the sand mold-forming composition; a step A2 for filling a molding space with the sand mold-forming composition in a foamed state; and a step A3 for solidifying the sand mold-forming composition filled in the molding space.
- 12.** A method for producing a sand mold, comprising: a step B1 for preparing the sand mold-forming composition of claim 8 by mixing and foaming the sand mold-forming additive and the silicate compound (C) and then adding the sand thereto; a step B2 for filling a molding space with the sand mold-forming composition; and a step B3 for solidifying the sand mold-forming composition filled in the molding space.
- 13.** A sand mold, comprising: the sand mold-forming additive of claim 1, sand, and a silicate compound (C) wherein the silicate compound (C) comprise at least one selected from sodium silicate, potassium silicate, and lithium silicate.
- 14.** The sand mold-forming additive of claim 5, wherein the surfactant (A) is 1 to 40 parts by mass and the surfactant (B) is 60 to 99 parts by mass when a total content of the surfactant (A) and the surfactant (B) is 100 parts by mass.
-